

Problem Statement

PSRO-style methods scale poorly in multi-agent reinforcement learning because they require explicit policy populations and full payoff matrices, causing high memory and computation overhead.

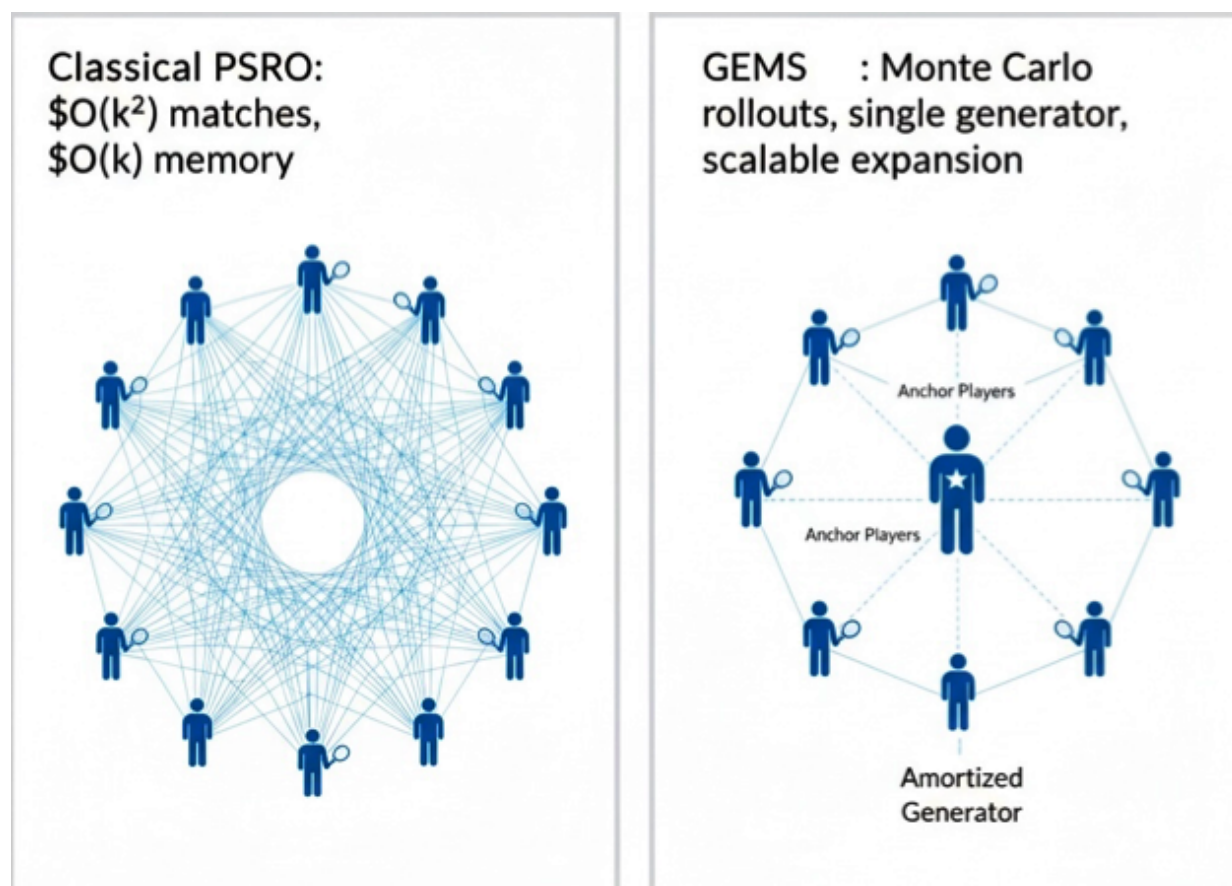


Fig.1 GEMS replaces PSRO's explicit policy populations and dense payoff-matrix construction with a compact generative framework, enabling scalable multi-agent learning with lower memory and computation overhead.

Proposed Framework

We propose **GEMS (Generative Evolutionary Meta-Solver)**, a compact and scalable framework that replaces explicit policy populations with:

- a **single amortized generator**,
- a **latent anchor set** representing active strategies,
- **Monte Carlo rollouts** for payoff estimation,
- **OMWU** for meta-strategy updates,
- **EB-UCB** for selecting promising new anchors,
- and **ABR-TR** for best-response training within the generator.

This design avoids explicit payoff-matrix storage and enables efficient policy expansion without training and storing separate actors for each new strategy.

Results

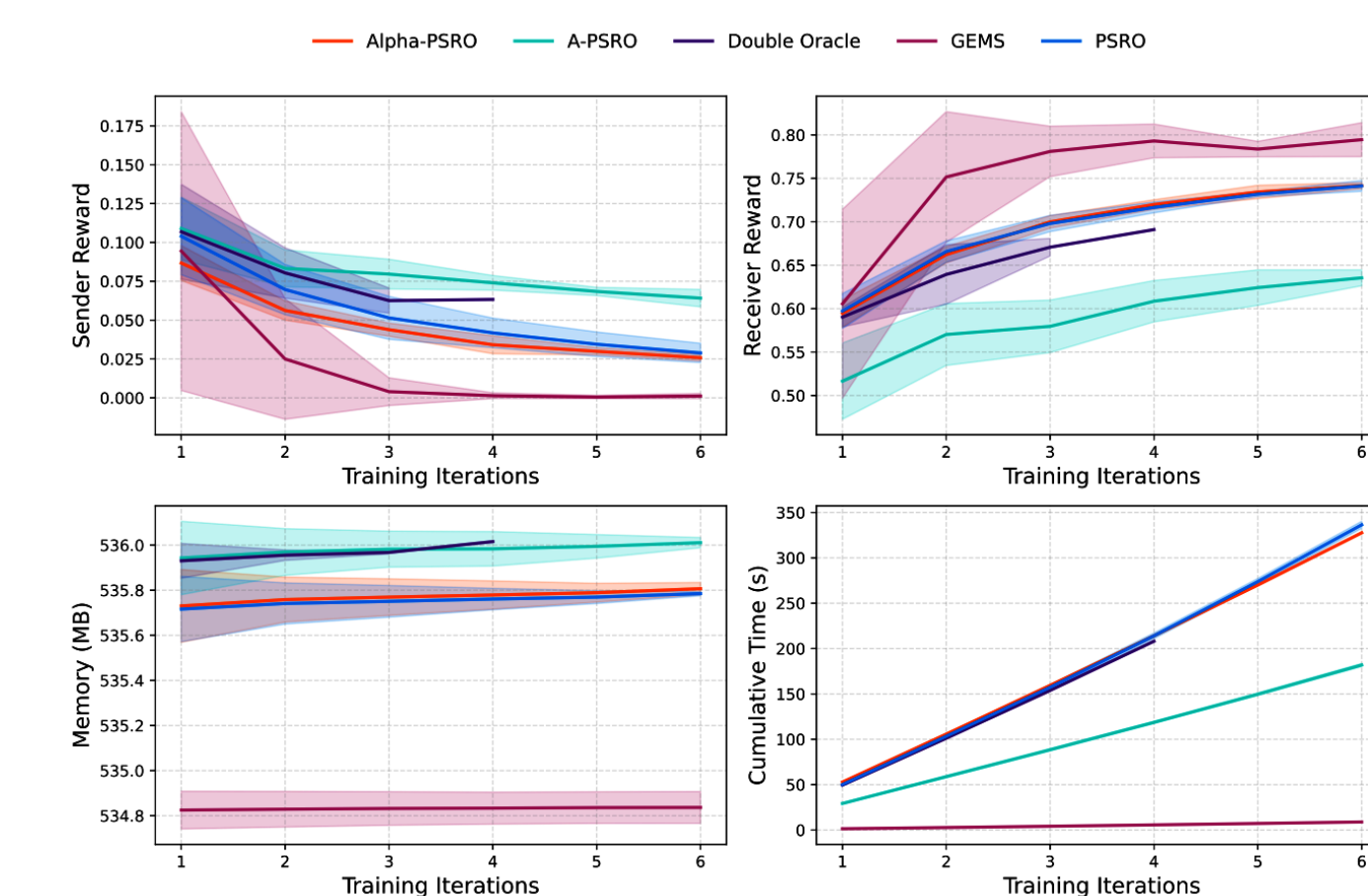


Fig.3 GEMS reaches the strongest equilibrium, driving sender reward near zero and receiver reward to its optimum with lower runtime and stable memory.

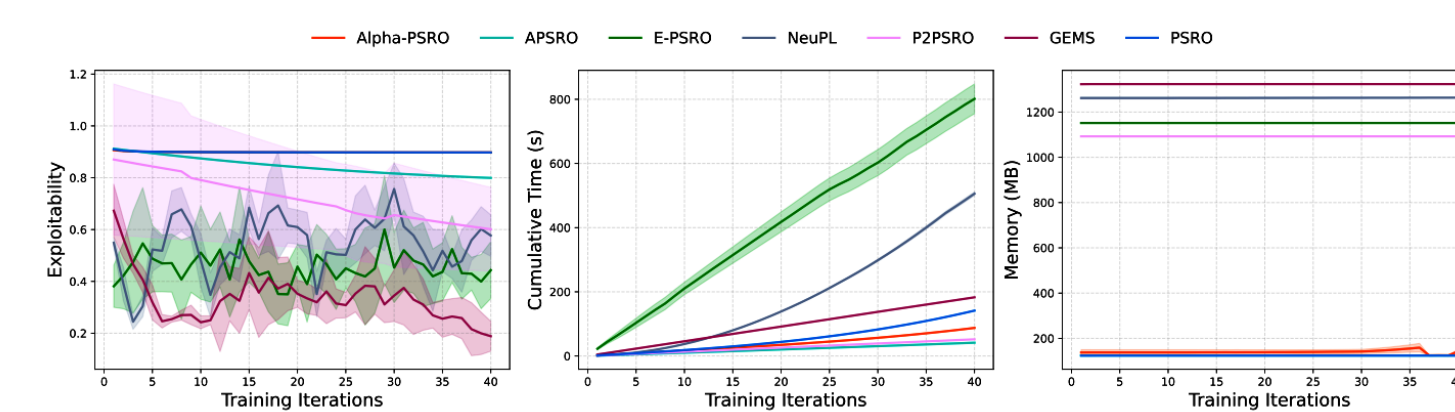


Fig.4 Kuhn Poker, GEMS achieves the lowest exploitability with better time and memory efficiency than strong PSRO baselines.

Research Gaps

- Existing PSRO-style methods improve best-response training and evaluation efficiency, but still rely on explicit policy populations and meta-game representations.
- As a result, they remain fundamentally limited in scalability for large and complex multi-agent environments.

Contributions

- 1. Memory-efficient policy representation**
GEMS replaces a growing population of separate policies with a single generator and a compact anchor set.
- 2. Computation-efficient meta-game estimation**
Instead of constructing the full payoff matrix, GEMS uses unbiased Monte Carlo rollouts to estimate policy values.
- 3. Scalable policy expansion**
EB-UCB identifies strong candidate strategies in latent space, while ABR-TR incorporates them without adding new actors.
- 4. Theoretical grounding**
GEMS provides guarantees through unbiased meta-estimates, regret bounds for oracle selection and OMWU updates, and finite-population exploitability analysis.

Activity Sequence Diagram

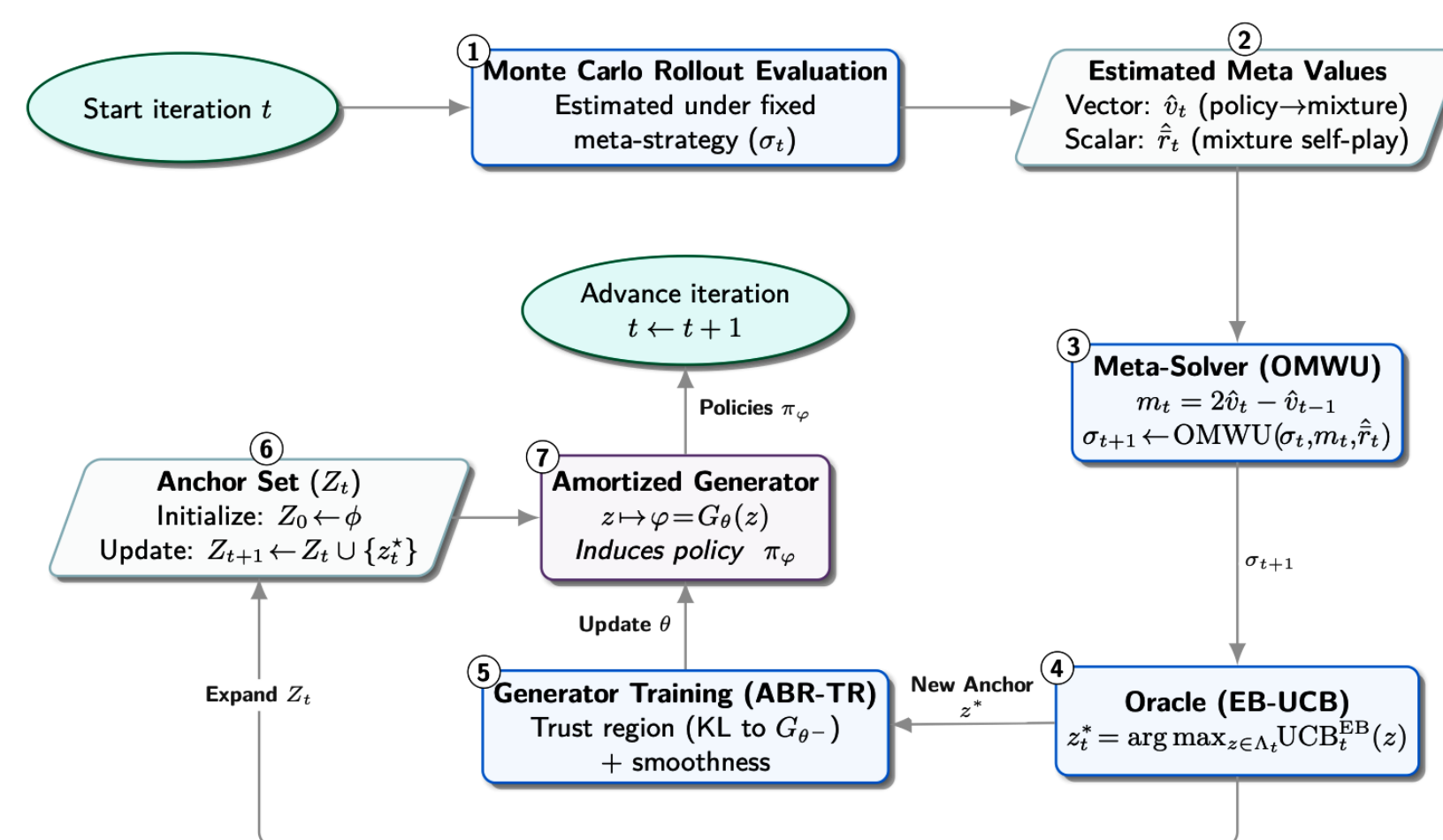


Fig.2 At each iteration, GEMS estimates meta-game values via Monte Carlo rollouts, updates the meta-strategy with OMWU, selects a promising latent anchor using EB-UCB, and trains the generator with ABR-TR to expand the policy set efficiently.

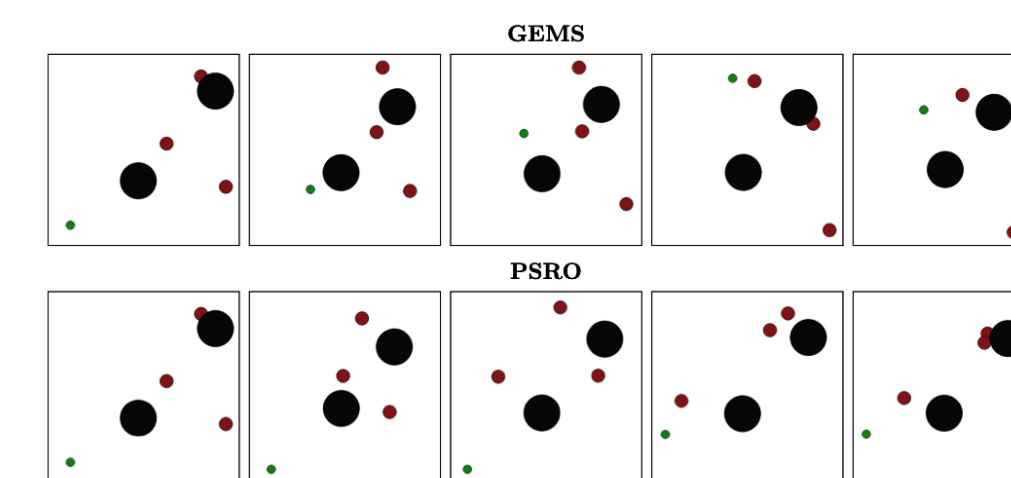


Fig.5 GEMS learns coordinated flanking and cornering behaviors, while PSRO shows simpler herding strategies.

References

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